



SIMATS ENGINEERING

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Course Code: DSA0603

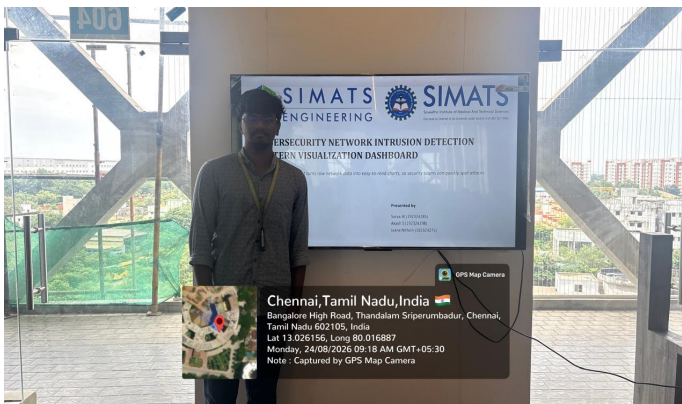
Slot: B

Course Name: Data Handling and Visualization

Course Faculty: Dr. Senthivadivu. S

Project Title: CYBERSECURITY NETWORK INTRUSION DETECTION PATTERN VISUALIZATION DASHBOARD

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

In this project, I developed an **Interactive Cybersecurity Threat Hunting Dashboard** that models multi-dimensional network traffic patterns as a statistical landscape, where connections are continuously evaluated for behavioral anomalies and known attack signatures. My primary contribution is the engineering of a predictive intrusion detection engine that fuses a **Random Forest classifier** with bootstrapped 95% confidence intervals, ensuring rigorous statistical bounding of attack-type distributions and automated outlier detection. This backend logic is integrated into a full-stack application using Python (Streamlit, Scikit-learn, Pandas), featuring a dynamic **NSL-KDD** data ingestion pipeline, Explainable AI (SHAP) integration, and an interactive interface that demonstrates instant, interpretable threat classification in response to user-simulated network payloads.

Student Signature

Guide Signature